

Spatial Degradation of Hepatocyte Zonation

Decoding the collapse of liver cell identity across the fatty-liver disease course



What the liver does — and the zonation gradient

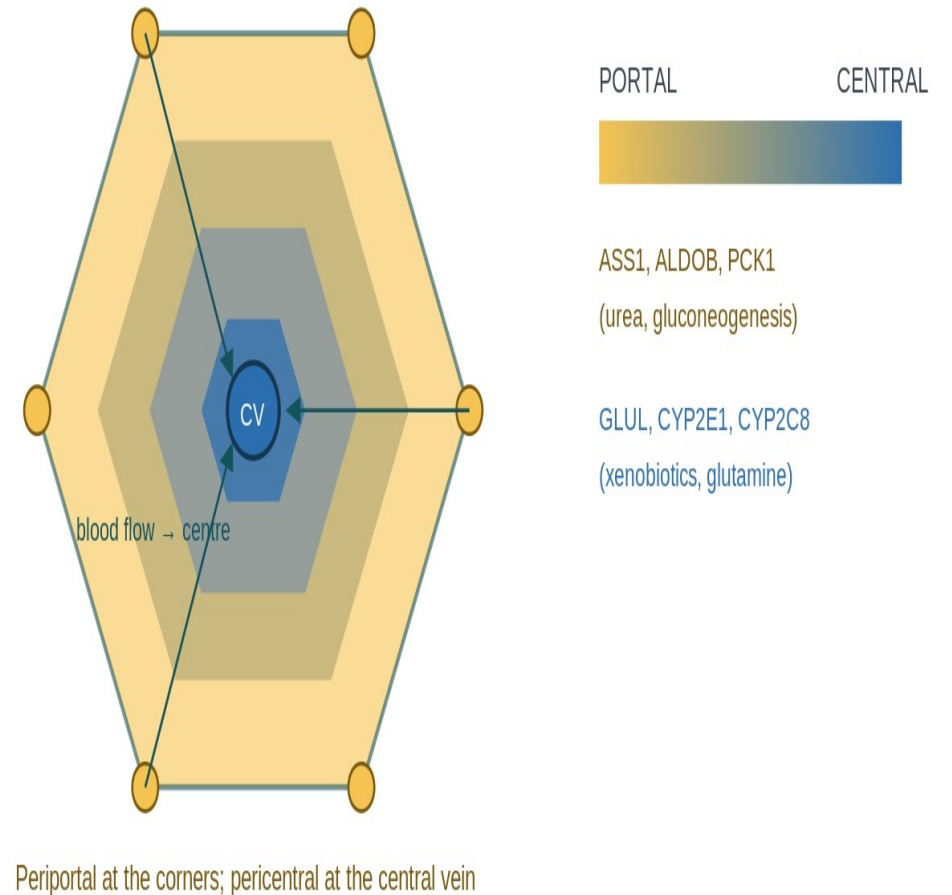
The liver is a metabolic factory: it buffers blood sugar, makes plasma proteins (albumin), produces bile acids, turns nitrogen waste into urea, and detoxifies drugs.

Two epithelial cells: hepatocytes (the bulk factory) and cholangiocytes (bile-duct lining).

Blood flows portal → central, so a cell's environment — and the genes it switches on — depends on where it sits:

- **Periportal:** urea cycle, gluconeogenesis (ASS1, PCK1).
- **Pericentral:** drug metabolism, glutamine (CYP2E1, GLUL).

The hinge: position is encoded in expression, so it can be decoded back out.



Two ways identity breaks: de-zonation vs plasticity

Both look like 'a cell expressing two opposite things at once' — the difference is WHICH boundary is crossed.

De-zonation

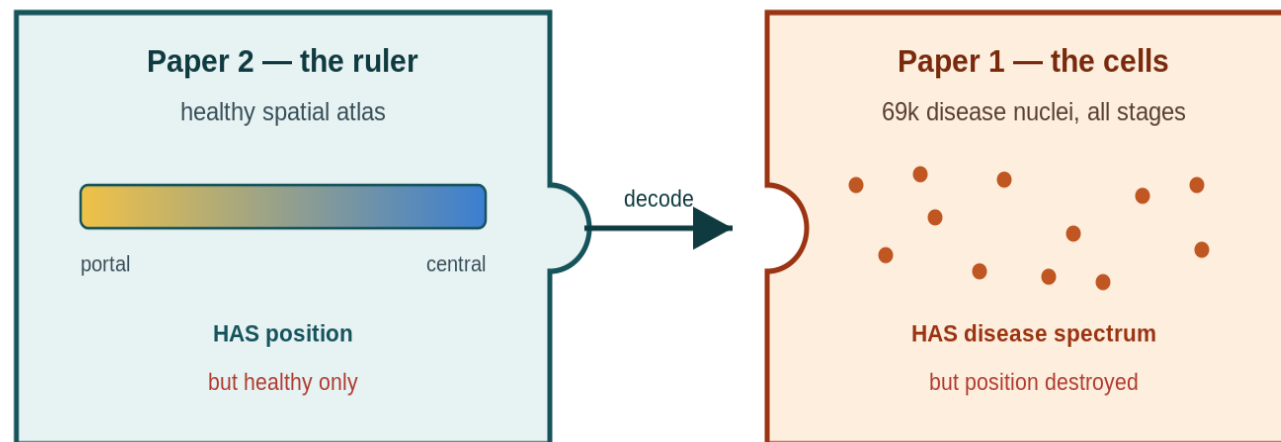
Loses its POSITION on the portal ↔ central axis (still a hepatocyte).
Co-expresses opposite zonal programs (CYP2E1 + ASS1).
Signature: the pericentral / periportal zonation genes.

Plasticity

Loses its CELL-TYPE identity (hepatocyte → bile-duct).
Co-expresses hepatocyte + duct programs (ALB + KRT7).
Signature: KRT7, KRT19, SOX9 ... (from Paper 1).

Analogy: a worker who forgets which station they man (de-zonation) is still a factory worker; one who forgets they work in the factory at all (plasticity) is something else. **H3 asks whether these are the same cells — does losing your address predict losing your job?**

Two datasets, one gap



Paper 2 — the ruler

Healthy human liver with spatial coordinates KEPT → a quantitative porto-central zone index. But healthy only.

Paper 1 — the cells

69k disease nuclei across all stages — but snRNA-seq THREW AWAY position.

Our move: use the healthy ruler to assign each diseased cell a pseudo-zonal coordinate. Neither paper did this.

Research question & hypotheses

Q Using the healthy zonation ruler, can we measure how — and how much — hepatocyte zonation degrades as MASLD advances from healthy to end-stage, and is that degradation linked to plasticity?

H1 — Erosion

A de-zonation score rises monotonically
Healthy → end-stage.

Test: ordered trend on per-donor values.

H2 — Which programs

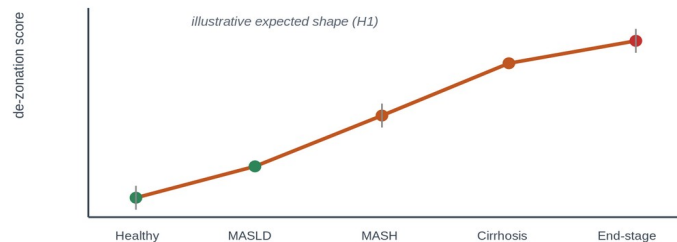
Specific programs (e.g. pericentral CYP detox)
collapse earliest.

Test: pseudobulk donor $\times$ zone DE + FDR.

H3 — Plasticity link

The most de-zonated cells are the ones gaining
bile-duct identity.

Test: within-donor correlation.



illustrative — the H1 collapse curve we're after

Why it matters: drug dosing (pericentral CYP) • ammonia / encephalopathy (periportal urea) • finer-than-histology staging • cancer / de-differentiation link • a reusable decoder for spatially-blind atlases.

How we decode position

1 Signatures

pericentral & periportal gene sets —
landmark & full, compared

2 Coordinate

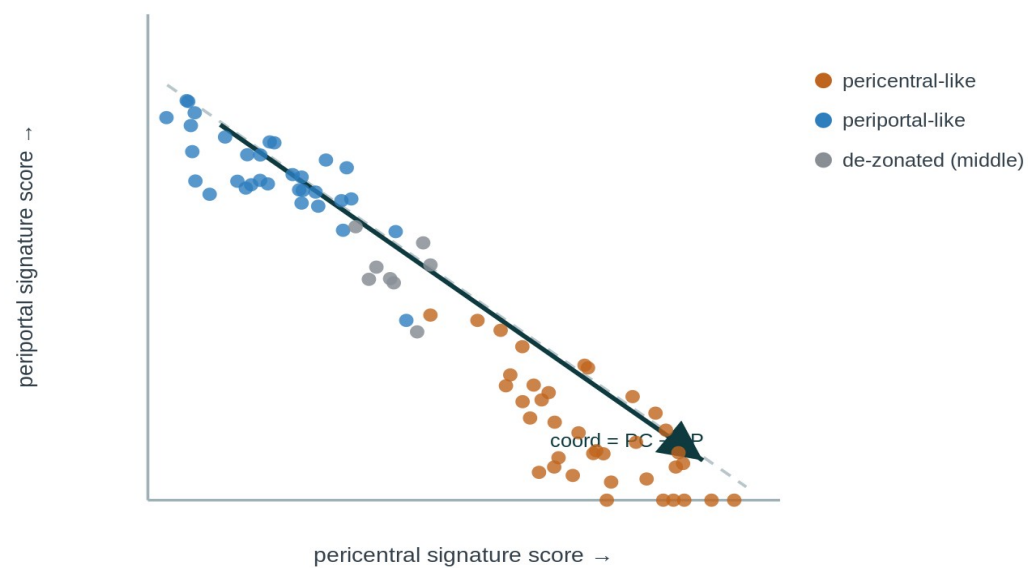
score = $\text{mean_z(PC)} - \text{mean_z(PP)}$ per
cell

3 Classifier

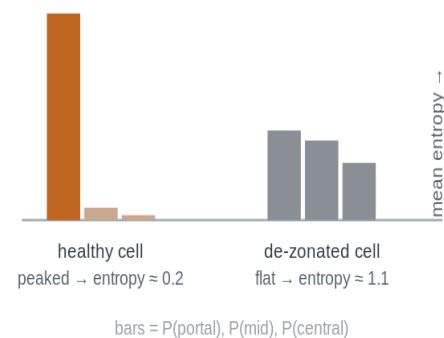
zone probabilities $\rightarrow$ entropy = confusion
= de-zonation

4 Collapse

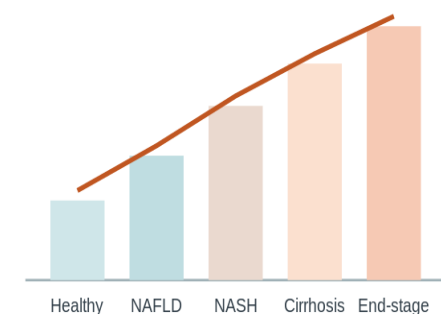
one metric per donor, tested across
stages



per-cell zone probabilities



mean entropy by stage (the meter)



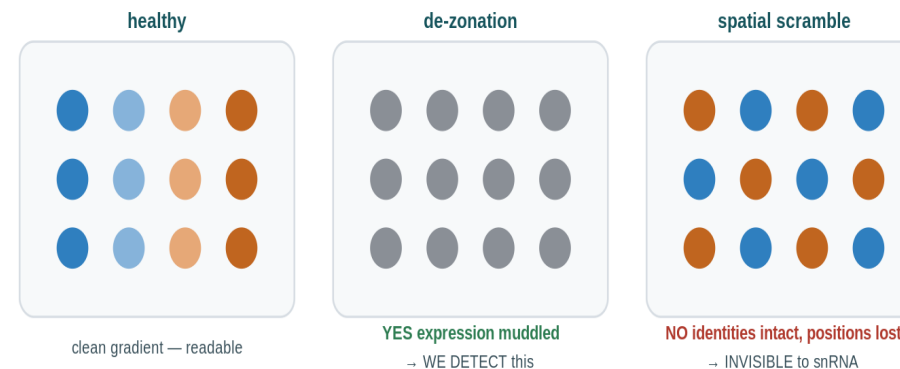
Doing it honestly

Negative control. Shuffle the disease-stage labels and re-run: the collapse trend must VANISH. A 'signal' that survives shuffling was an artefact. (The positive control — recovering zonation in healthy liver — is the next slide.)

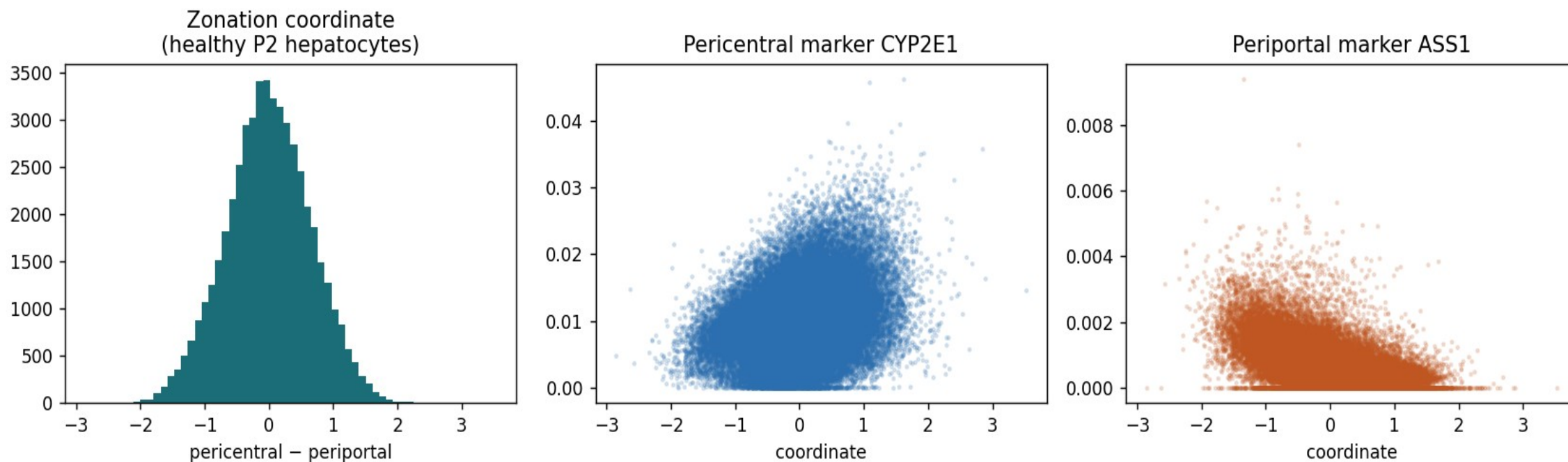
What de-zonation is — and isn't

IS — loss of the expression program: pericentral-periportal mutual exclusivity weakens and the coordinate spread narrows.

ISN'T — physical spatial scrambling with intact expression: dissociated data couldn't see it, and it's very unlikely here.



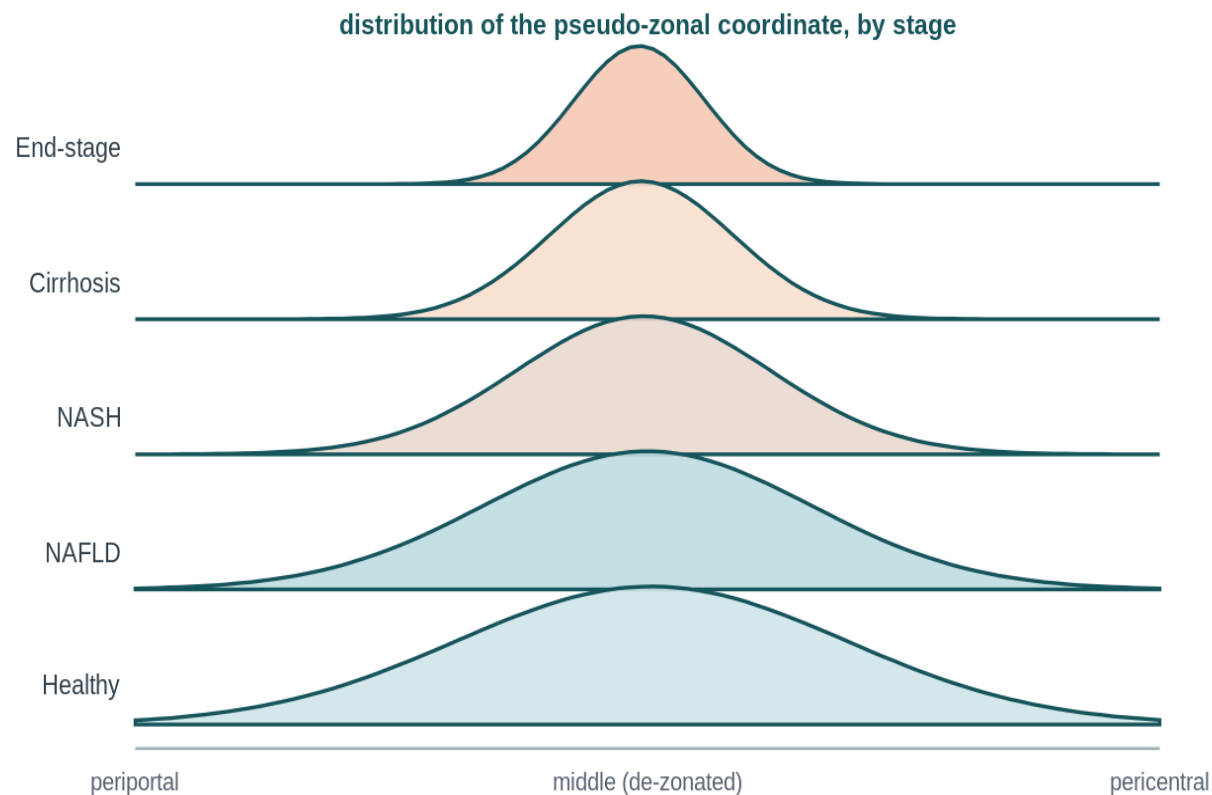
The ruler works on healthy liver



On real Paper 2 healthy hepatocytes the coordinate recovers zonation: pericentral CYP2E1 rises with it and periportal ASS1 falls.

A single broad central hump is exactly what a continuum looks like — the proof of zonation is the marker slopes, not two peaks. We validated before trusting it on disease.

H1 — Does zonation collapse across stages?



EXPECTED SHAPE — swap for real A6

What to look for

Coordinate spread **NARROWS** and pericentral-periportal anti-correlation weakens toward 0 across stages.

We found: ____

[fill: ρ , donor-bootstrap 95% CI, permutation p]

plot_a6_collapse · pipeline.py:collapse

H2 — Which programs collapse, and where?

FIGURE GOES HERE

Volcano per zone (portal / central): effect vs $-\log_{10} q$

plot_a7_volcano · signature genes flagged

What to look for

A coherent set of genes at $q < 0.05$ — surviving EXCLUSION of the signature genes — concentrated in one zone first.

We found: ____

[fill: # genes $q < 0.05$ per zone; name the 3–5 earliest-collapsing programs]

pipeline.py:de

H3 — Are de-zonated cells the plastic ones?

FIGURE GOES HERE

Stage-stratified scatter: de-zonation vs plasticity score

plot_a8_plasticity · WITHIN donor/stage

What to look for

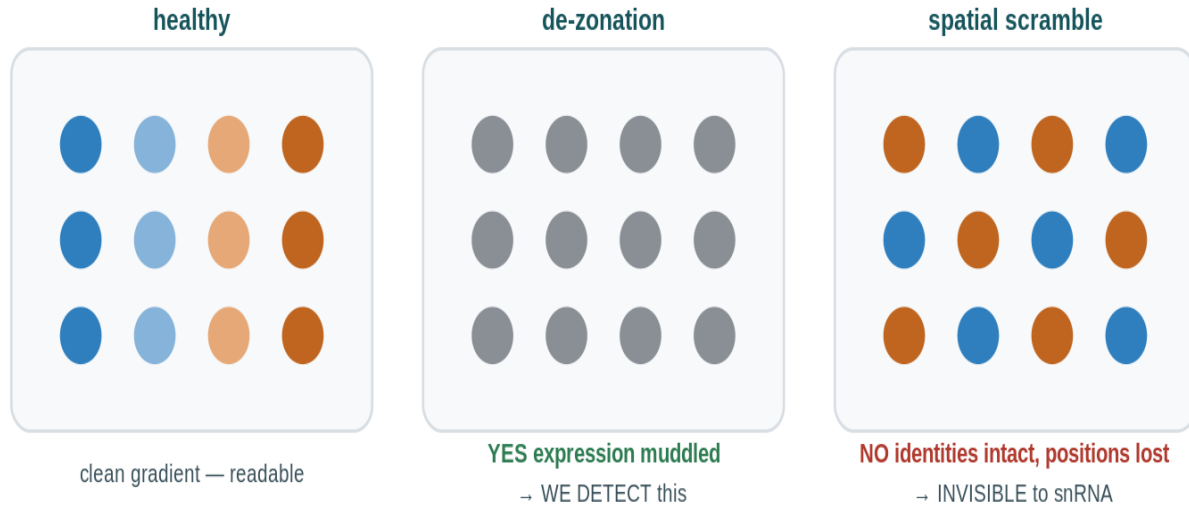
A positive within-donor link — not a pooled one (late disease trivially has more of both).

We found: ____

[fill: mean per-donor ρ ; OLS dez coefficient controlling stage+donor]

pipeline.py:plasticity

Limitations & honest scope



- **Blind to pure spatial scrambling** — if cells keep identity but move, dissociated data sees nothing.
- **Pseudo-zonal, not (x,y)**
 - a 1-D reconstructed coordinate, never a true position.
- **Ruler-validity matters**
 - disease may corrupt the marker genes; Step 5b checks the ruler still holds.
- **Associational**
 - we measure collapse, not its cause.

Conclusions

H1 — [did zonation collapse, and how monotonically? ____]

H2 — [which programs eroded first, and in which zone? ____]

H3 — [did de-zonation track plasticity, within donors? ____]

One-line takeaway: a reference-decoder turns the many spatially-blind disease atlases into a quantitative readout of how liver identity disintegrates. [refine with results]

Thank you — questions?

Roei · Shira | Computational Genomics 76553 (backup slides follow)

Is the ruler still a ruler? (validity battery)

FIGURE GOES HERE

Per-stage: internal coherence · cross-program anti-corr · split-half · program-off vs restriction-lost

plot_a5b_ruler (Step 5b)

Classifier — held-out confusion matrix (Paper 2)

FIGURE GOES HERE

Confusion matrix + held-out accuracy on Paper 2 before applying to Paper 1

Step 4b · calibrated multinomial logistic regression

Statistical rigor — the checklist

- BH-FDR $q < 0.05$, never raw p
- Ordered trend test, not pairwise only
- Donor-level resampling (never 69k cells as independent)
- Pseudobulk donor $\times$ zone DE; H2 via coordinate $\times$ stage interaction on held-out genes (repeated random splits)
- Positive + negative controls
- Effect sizes beside p/q

Bonus — collapse vs inherited risk (Paper 3)

FIGURE GOES HERE

Hypergeometric overlap: Step-7 collapse drivers vs Paper-3 risk-variant targets, expression-matched background

step9_bonus_enrichment.py · report fold + BH q

Data & reproducibility

Raw → processed: Seurat .rds → counts.mtx; MATLAB .mat → paper2_train.npz. Pipeline reads only data/processed/.

Repo:

src/{prep,steps,plotting,tools}, signatures/{full,paper2_landmark,core,expanded}, results/. run_all.py reproduces every figure on a clean checkout.

Signatures:

reported BOTH ways — the full transcriptome-wide program (default) and the exact Paper 2 landmark set — as a robustness comparison.